

Agniveer Vayu Other than Science (Group Y) - 19 Jan 2023 - Memory Based Paper

English

Question 1

Some parts of the sentence have errors and some are correct. Find out which part has an error and mark that part as your answer. If there are no errors, mark 'No error' as your answer.

Hardly had he departed that the phone rang.

Options:

- A. Hardly had
- B. he departed that
- C. the phone rang
- D. No error

Answer: B

Solution:

The erroneous part is "he departed that"



Key Points

- In Option 2, the usage of *"that" is incorrect instead of this we need to use "when"*.
- With **"hardly/scarcely"** we always use a conjunction **"when"**.
- Let's see an example-
 - **Hardly** had I reached the station **when** the train came.
- Hence, the correct sentence is- *Hardly had he departed **when** the phone rang.*

Question 2

Identify the part of the sentence which has an error in it. In case there is no error the answer will be (D).

Ram has an (A)/ large amount(B)/ of money. (C)/ No error (D)

Options:

A. A

B. B

C. C

D. D

Answer: A

Solution:

The correct answer is "A".



Key Points

- In the given sentence, the usage of "an" is incorrect. Instead of "an", we need to use "a".
- We use "an" with nouns which have vowel sound and we use "a" with nouns which have consonant sound. "large" has consonant sound. Hence, we need to use "a".
- The correct sentence is: "Ram has *a large amount of money*".



Additional Information

- *Although 'Number and amount' have similar meanings, there is a difference between them:*
 - 'Number' is used for things that can be counted, while 'Amount' is used for things that cannot be counted.
- For example:
 - The **number** of butter was not enough for him. ❌
 - The **amount** of butter was not enough for him. ✅

Question 3

Directions: Each item in this section has a sentence with three underlined parts labelled as (a), (b) and (c). Read each sentence to find out whether there is any error in any underlined part and indicate

your response on the answer sheet against the corresponding letter, i.e., (a) or (b) or (c). If you find no error, your response should be indicated as (d).

They had to wait for the luggages (a)/ to be put into the cab (b)/ before they themselves got into it(c)/ No error(d).

Options:

A. (a)

B. (b)

C. (c)

D. (d)

Answer: A

Solution:

The correct answer is "They had to wait for the luggage".

Key Points

- There are some **uncountable nouns** (followed by a singular verb) which don't take the articles 'a/an', many, few, and plural forms:
 - Furniture, scenery, luggage, temper, hair, knowledge, equipment, pottery, music, jewellery, etc.
 - Example:
 - The scenery at my friend's house enchanted me.* ❌
 - The scenery at my friend's house enchanted me.* ✅
- As per the rule given above, 'luggage' will be used instead of 'luggages' in the 4th part of the given question.

Correct sentence: They had to wait for the **luggage** to be put into the cab before they themselves got into it.

Additional Information

- The nouns given below seem plural in form but in actuality are singular:
 - news, billiards, summons, physics, measles, innings, etc.
 - Example:
 - Physics is an interesting subject.*
- हाँ कुछ **uncountable nouns** (followed by a singular verb) जो article 'a/an', many, few, and plural form: नहीं लेते हैं:
 - Furniture, scenery, luggage, temper, hair, knowledge, equipment, pottery, music, jewellery, etc.
 - उदाहरण:

▪ *The **sceneries** at my friend's house enchanted me.* ❌

▪ *The **scenery** at my friend's house enchanted me.* ✅

- ऊपर दिए गए नियम के अनुसार, 'luggage' को देखते हुए सवाल के 4th part में 'luggages' के स्थान पर इस्तेमाल किया जाएगा।

सही वाक्य: *They had to wait for the **luggage** to be put into the cab before they themselves got into it.*

Question 4

Select the option that can be used as a one-word substitute for the given group of words.

A decision on which one cannot go back

Options:

- A. Irrevocable
- B. Improbable
- C. Incredible
- D. Incorrigible

Answer: A

Solution:

The correct answer is 'Irrevocable'.



Key Points

- Let us look at the meaning of the given options -
 - **Irrevocable** -not able to be changed, reversed, or recovered; final.
 - **Improbable** -not likely to be true or to happen.
 - **Incredible** -impossible to believe.
 - **Incorrigible** -(of a person or their behaviour) not able to be changed or reformed.
 - From the above meanings it is clear that **irrevocable** is the best word to substitute the given group of words.
 - Hence, **option 1** is the correct answer.
-

Question 5

Choose the correctly spelt word.

Options:

- A. Peirce
- B. comfortable
- C. Luxuries
- D. Indeividual

Answer: B

Solution:

The correct answer is '**comfortable**'.

Key Points

- **Comfortable** is spelt correctly. It means (especially of clothes or furnishings) providing physical ease and relaxation.
 - The correct spellings (and the meanings) of the other words are:
 - **Peirce**--Pierce: to make a hole in something with a sharp point. (नुकीले औज़ार से किसी वस्तु में छेद करना)
 - **Example**--*The needle pierced into her skin.*
 - **Luxuries**--Luxuries: indulgence in and enjoyment of rich, comfortable, and sumptuous living. (विलासिता)
 - **Example**--*The life of the ruling class was comfortable and even luxurious from early times.*
 - **Indeividual**--Individual: single/ separate. (समूह से अलग; अकेला)
 - **Example**--*These styles can be adapted to suit individual tastes.*
 - Hence, the correct answer is **option 2**.
-

Question 6

Select the most appropriate synonym of the given word.

STERN

Options:

- A. strict
- B. mild

C. lenient

D. gentle

Answer: A

Solution:

The correct answer is '**strict**'.

Key Points

- The most appropriate synonym of the given word '**Stern**' is **Strict**'.
- **Stern**: very serious; not smiling. (कठोर, बहुत गंभीर; मुस्कान-रहित)
 - *Example: His voice was **stern** again.*
- **Strict**: not allowing people to break rules or behave badly. [सख्त, कठोर (नियम भंग या गलत व्यवहार को सहन न करने वाला)]
 - *Example: Samir's very **strict** with his children.*

Correct Answer: *strict.*

Additional Information

- Let's look at the **meaning** of other words:
 - **mild**: not strong; not very bad. (नरम; हलका)
 - **lenient**: (used about punishment or person who punishes) not as strict as expected. [(दंड या दंड देने वाला व्यक्ति) जो आशा के अनुरूप कठोर न हो; नरम, सौम्य]
 - **gentle**: (used about people) kind and calm; touching or treating people or things in a careful way so that they are not hurt. [(व्यक्ति) दयालु और शांत; बिना चोट पहुँचाए कोमलता से व्यवहार करने वाला; सौम्य]
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Question 7

Direction: In the following questions, choose the word opposite in meaning to the given word.

Condemn

Options:

A. Castigate

B. Strut

C. Command

D. Commend

Answer: D

Solution:

The correct answer is 'Commend'.



Key Points

- The word 'Condemn' is a verb and it means **to criticize something or someone strongly, usually for moral reasons.**
 - For example, *The 26/11 terrorist attack in Mumbai was **condemned** as an act of barbarism and cowardice.*
- The marked option 'Commend' means **to formally praise someone or something.**
 - For example, *The judge **commended** the girl for her bravery.*
- So, we can say that '**Commend**' can function as the opposite word of '**Condemn**.'

Hence, the correct answer is **Option 4**.



Additional Information

- **Castigate** ⇒ to criticize someone or something severely.
- **Strut** ⇒ to walk in a proud, confident gait.
- **Command** ⇒ to order someone (esp in military) or control and responsibility over something.

Question 8

Direction: Choose the most suitable answer to fill the blank:

If you had told me you needed a ride, I _____ earlier.

Options:

A. would have left

B. would have been left

C. will have left

D. will be leaving

Answer: A

Solution:

The correct answer is '**would have left.**'

Key Points

- The given sentence is the **type 3 conditional**.
- The **type 3 conditional** is used to refer to a time that is in the past, and a situation that is contrary to reality.
- The structure of **type 3 conditional** is:

If clause	Main clause
If + past perfect	perfect conditional or perfect continuous conditional

- **Example:** *If it had rained you would have gotten wet.*
- Hence, the correct answer is '**would have left.**'
- **Correct sentence:** *If you had told me you needed a ride, I **would have left** earlier.*

Additional Information

- The **zero conditional** is used for when the time being referred to is now or always and the situation is real and possible.
- The **zero conditional** is often used to refer to general truths.
- The tense in both parts of the sentence is the simple present.
- In zero conditional sentences, the word "**if**" can usually be replaced by the word "**when**" without changing the meaning.

Hinglish

- दिया गया वाक्य **type 3 conditional** है।
- **Type 3 conditional** का उपयोग उस समय को संदर्भित करने के लिए किया जाता है जो अतीत में है, और ऐसी स्थिति जो वास्तविकता के विपरीत है।
- **Type 3 conditional** की structure है:

If clause	Main clause
If + past perfect	perfect conditional or perfect continuous conditional

- **Example:** *If it had rained you would have gotten wet.*

Question 9

Direction: Choose the most effective word for the blank from the options given below.

Hardly had I reached my office _____ I got your message.

Options:

- A. than
- B. before
- C. after
- D. when

Answer: D

Solution:

The correct answer is 'when.'



Key Points

- If the second event occurs immediately after the first, we can express that idea using the structure **hardly ... when**.
 - यदि पहली के तुरंत बाद दूसरी घटना होती है, तो हम उस विचार को संरचना **hardly ... when** का उपयोग करके व्यक्त कर सकते हैं।
 - **Example:** *Hardly had I reached the station **when** the train came.*
- **Correct sentence:** *Hardly had I my office **when** I got your message.*



Additional Information

- When we begin a sentence with a negative word, we put the auxiliary verb before the subject.
 - **Example:** *No sooner had she read the letter than she started crying.*
~~*No sooner she read the letter than she started crying.*~~

Question 10

Choose the option with the appropriate question tag for the given sentence:

George has never been to India, _____

Options:

- A. has he?

- B. has George?
- C. hasn't he?
- D. hasn't George?

Answer: A

Solution:

The correct answer is '**has he?**'.

Key Points

- A **question tag** means a short phrase such as 'isn't it?' or 'did you?' at the end of a sentence that changes it into a question and is often used to ask somebody to agree with you.
 - For example: *She said half price, **didn't she?***
- If the **main clause** is **positive**, the **question tag** is usually **negative**, and if the **main clause** is **negative**, the **question tag** is **positive**.
- The sentence and the question tag must be in the **same tense**.
- For the **negative question tag**, use the **Contracted form of 'helping verb'** and '**not**'
- By referring to the above explanation, it can be concluded that **has he?** should be the appropriate option to be chosen for filling the blank.

Hence, the correct answer is **option 1**.

Question 11

Direction: Choose the option that best transforms the sentence into its other voice:

We added up the money and found it was correct.

Options:

- A. Adding up the money it was found it to be correct by us.
- B. Adding up the money we found it to be correct.
- C. The money was added up and found to be correct.
- D. The money is added up by us and found to be correct.

Answer: C

Solution:

The correct answer is 'The money was added up and found to be correct.'

Key Points

- The given sentence is in **Active Voice**. As per the given question we have to change it into **Passive Voice**.
- The structure of the given sentence is as follows:
 - **Subject + V2 + Object. (Active Voice)**
 - **Subject (objective case) + was/were + V3 + Object (subjective case). (Passive Voice)**
- **Example:**
 - *Their cousin showed them some rare photographs. (Active Voice)*
 - *They were shown some rare photographs by their cousin. (Passive Voice)*
- In the first part of the sentence:
 - **The subject** of the given sentence is '**We**'.
 - **The object** of the given sentence is '**The money**'.
 - **The subject** will be put in place of **the object** and **the object** will be put in place of **the subject**.
 - And '**added up**' will be changed into '**was added up**'.
 - '**Money**' must use '**was**' with it.
- As the second part is connected with the help of the conjunction "**and**" so the same rules will be applied to the verb "**found**."
- Therefore, as per the points mentioned above, we find that the correct answer is **Option 3**.

Correct sentence: *The money was added up and found to be correct.*

Question 12

Direction: Choose the option that best transforms the sentence into its direct / indirect form:

Can the children get tasty meals at school?

Options:

- A. I asked if the children can be getting tasty meals at school.
- B. I asked if the children could get tasty meals at school?
- C. I asked if the children can they get tasty meals at school?
- D. I asked if the children could get tasty meals at school.

Answer: D

Solution:

The correct answer is 'I asked if the children could get tasty meals at school.'

Key Points

- The given interrogative sentence is in the **direct speech**.
- The rules of **Direct and Indirect speech** for **interrogative sentence** conversions.
 - Inverted commas are removed in indirect speech.
 - The reporting verb said/said to is changed to "**asked, demanded, ordered, enquired**" as per the nature of the sentence.
 - The first person in the reported speech changes according to the subject of reporting speech.
 - In Reported Speech change of the second person depends on the object of reporting speech.
 - The third person remains unchanged.
 - To make the indirect speech for questions that can be answered in YES/NO, the word "**if**" or "**whether**" is used before the reported speech within the sentence.
 - "**Can**" changes to "**could**."
 - **Structure:**
 - (Can + Subject+ V1 + Object?) (Direct Speech)
 - (If + Subject + could + V1 + Object) (Indirect Speech)
 - **Example:**
 - He said to me "Can you help me?"
 - He asked me if I could help him.
- Therefore, the correct answer is **Option 4**.

Correct sentence: *I asked if the children could get tasty meals at school.*

Confusion Points

- **Option 2** and **4** are different.
 - There is a difference in punctuation marks.
 - One option has a question mark and the other has a full stop.
-

Question 13

Arrange the parts(1), (2), (3) and (4) of the following sentences to create meaningful sentences and choose the right option accordingly.

1) people think they're

2) Blackberry phones used to be

3) old-fashioned gadgets

4) extremely popular, but now many

Options:

A. 4, 3, 1, 2

B. 3, 1, 2, 4

C. 4, 1, 3, 2

D. 2, 4, 1, 3

Answer: D

Solution:

The correct answer is '2, 4, 1, 3'.

Key Points

- The part '2' is independent of any other sentences as it is giving information about the **noun** '*Blackberry phones*' that *earlier they used to very popular but with the passage of time, people are considering them old fashioned gadgets*. Hence, '2' is the **first part**.
- The **phrase** '*extremely popular*' mentioned in the part '4' refers back to the **noun** '*Blackberry phones*' mentioned in the part '2'. Hence, '4' follows '2'.
- The **noun** '*people*' mentioned in the part '1' is linked with the **phrase** '*but now many*' mentioned in the part '4'. Hence, '1' follows '4'.
- The sentence '3' is the **concluding sentence**. Hence, '3' is the **last sentence**.
- The correct sequence is **2, 4, 1, 3**.

Paragraph after rearranging the sentences: *-Blackberry phones used to be extremely popular, but now many people think they're old-fashioned gadgets.*

Hence, the correct answer is **option 4**.

Question 14

Given here are four jumbled sentence. Pick the option that gives the correct order.

A. One day, Maytag's company will create a product

B. which will wash 660 million loads of laundry per week

C. and become an empire worth 1.7 billion dollars,

D. found in over 80 per cent of American households.

Options:

A. BADC

B. BDCA

C. ABDC

D. ABCD

Answer: D

Solution:

The correct answer is "ABCD"



Key Points

- The first sentence will be A because it introduces Maytag's company and its product
 - **One day, Maytag's company will create a product**
- The second sentence will be B because it states further about the utility of the product
 - **which will wash 660 million loads of laundry per week**
- The third sentence will be C because it states about the empire created by the product
 - **and become an empire worth 1.7 billion dollars.**
- The last sentence will be D because it describes the empire as so huge that the product will be found in every house.
 - **found in over 80 per cent of American households.**

Hence, the correct passage is: *One day, Maytag's company will create a product which will wash 660 million loads of laundry per week and become an empire worth 1.7 billion dollars, found in over 80 per cent of American households.*

Question 15

Select the noun form of the given word:

Different

Options:

A. Difference

B. Differ

C. Differently

D. Differentness

Answer: A

Solution:

The correct answer is "**Difference**"

Key Points

- The correct answer is "**Difference**"
- The given word "**Different**" is an adjective that means "**not the same as another or each other; unlike in nature, form, or quality.**"
- "**Difference**" is a noun that means "**the way in which two or more things which you are comparing are not the same**"
- Let's see examples-
 - She seems to wear something **different** every day.
 - What's the **difference** between an ape and a monkey?

Hence, the correct answer is "**Option 1 i.e. Difference**"

Question 16

Direction: Find out which part has an error and mark it as your answer. If there is no error, mark 'No error' as your answer.

No sooner did she see her teacher (A) / when she (B) / stopped copying. (C) / No error (D)

Options:

A. (A)

B. (B)

C. (C)

D. (D)

Answer: B

Solution:

The correct answer is '(B).'

Key Points

- The error lies in **Part B** of the sentence.
- In a sentence, the use of '**when**' is incorrect.
- The conjunction '**No sooner ---- than**' is used to denote simultaneous actions.
- '**No sooner**' is always followed by '**than**' and not '**when**.'
 - **Example:** *No sooner had I received her call **than** I left for her place.*
- Therefore, the use of '**when**' in **Part B** of the sentence should be replaced with '**than**.'
- Hence, the correct answer is '(B).'

The Correct Sentence: *No sooner did she see her teacher **than** she stopped copying.*

Additional Information

- A **conjunction** is a joiner, a word that connects (conjoins) parts of a sentence.
- **Coordinating conjunctions** join equals to one another.
 - **Example:** *Most children like sweets **and** cookies.*
- Coordinating conjunctions go in between items joined, not at the beginning or end.
 - **Example:** *I like coffee, **but** I don't like tea.*



- दिए गए वाक्य का हिंदी अनुवाद:
 - जैसे ही उसने अपने शिक्षक को देखा उसने नकल करना बंद कर दिया।
- एक वाक्य में '**when**' का प्रयोग गलत है।
- साथ-साथ होने वाली क्रियाओं को दर्शाने के लिए conjunction '**No sooner ---- than**' का प्रयोग किया जाता है।
- '**No sooner**' के बाद हमेशा '**than**' आता है न कि '**when**'।

Question 17

Directions: Read the following passage and answer the given questions:

Religion and science have often been viewed as having conflicting ideologies, with one relying on faith and the other on evidence. However, many argue that these two concepts can coexist and even complement each other. Religion can provide a moral framework and a sense of purpose, while science can explain the natural world and how it operates. One way in which religion and science can coexist is through the concept of intellect. Religion provides a moral framework and a sense of purpose, while science explains the natural world and how it operates. One example of this coexistence is in the study of the origins of the universe. Many religions have creation myths that describe how the world came to be, while science has the Big Bang theory. Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other.

In conclusion, religion and science are not mutually exclusive. They can coexist and even complement one another, providing a deeper understanding of the world and our place in it. Both offer unique perspectives and insights, and neither necessarily contradicts the other. The key is to approach each with an open mind and a willingness to consider different perspectives.

What is the relationship between religion and science?

Options:

- A. Religion and science have had a simple relationship throughout history.
- B. Science is a systematic method for understanding the natural world.
- C. Religion and science can coexist and even complement each other.
- D. Science is seen as a threat to traditional religious beliefs.

Answer: C

Solution:

The correct answer is "**Religion and science can coexist and even complement each other.**"

Key Points

- Option '3' is correct.
 - From the passage: '*.... Religion and science have often been viewed as having conflicting ideologies, with one relying on faith and the other on evidence. However, **many argue that these two concepts can coexist and even complement each other**.....*'
 - The passage states that **religion and science can coexist and even complement each other.** Religion provides a moral framework and a sense of purpose, while science explains the natural world and how it operates.
 - Thus, in this context, we can say that the correct answer is option "3"
- Options 1, 2, and 4 are incorrect because there isn't sufficient supportive information in the passage to consider them to be the right answers.

Hence, the correct answer is "*Religion and science can coexist and even complement each other.*"

Question 18

Directions: Read the following passage and answer the given questions:

Religion and science have often been viewed as having conflicting ideologies, with one relying on faith and the other on evidence. However, many argue that these two concepts can coexist and even complement each other. Religion can provide a moral framework and a sense of purpose, while science can explain the natural world and how it operates. One way in which religion and science can coexist is through the concept of intellect. Religion provides a moral framework and a sense of purpose, while science explains the natural world and how it operates. One example of this coexistence is in the study of the origins of the universe. Many religions have creation myths that describe how the world came to be, while science has the Big Bang theory. Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other.

In conclusion, religion and science are not mutually exclusive. They can coexist and even complement one

another, providing a deeper understanding of the world and our place in it. Both offer unique perspectives and insights, and neither necessarily contradicts the other. The key is to approach each with an open mind and a willingness to consider different perspectives.

What is the key to approaching religion and science according to the passage?

Options:

- A. Religion and science can both provide valuable insights and understanding.
- B. The key is to approach each with an open mind and a willingness to consider different perspectives.
- C. The key is to avoid conflating the two; religion and science are mutually exclusive.
- D. Many religious beliefs are not in opposition to scientific facts and theories.

Answer: B

Solution:

The correct answer is "The key is to approach each with an open mind and a willingness to consider different perspectives."



Key Points

- Option '2' is correct.
 - From the passage: '.... Both offer unique perspectives and insights, and neither necessarily contradicts the other. **The key is to approach each with an open mind and a willingness to consider different perspectives.....**'
 - According to the passage, the key to approaching religion and science is recognizing that they offer different perspectives on the same question and can coexist by considering them as valid ways of understanding and making sense of the world.
 - Thus, in this context, we can say that the correct answer is option "2"
- Options 1, 3, and 4 are incorrect because there isn't sufficient supportive information in the passage to consider them to be the right answers.

Hence, the correct answer is "*The key is to approach each with an open mind and a willingness to consider different perspectives.*"

Question 19

Directions: Read the following passage and answer the given questions:

Religion and science have often been viewed as having conflicting ideologies, with one relying on faith and the other on evidence. However, many argue that these two concepts can coexist and even complement each other. Religion can provide a moral framework and a sense of purpose, while science can explain the natural world and how it operates. One way in which religion and science can coexist is through the concept of intellect. Religion provides a moral framework and a sense of purpose, while science explains the natural world and how it operates. One example of this coexistence is in the study of the origins of the universe. Many religions have creation myths that describe how the world came to be, while science has the Big Bang theory. Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other.

In conclusion, religion and science are not mutually exclusive. They can coexist and even complement one another, providing a deeper understanding of the world and our place in it. Both offer unique perspectives and insights, and neither necessarily contradicts the other. The key is to approach each with an open mind and a willingness to consider different perspectives.

Select the most appropriate **ANTONYM** of the given word. "INTELLECT"

Options:

- A. Genius
- B. Wizard
- C. Stupidity
- D. Sage

Answer: C

Solution:

The correct answer is "Stupidity."



Key Points

- "Intellect" refers to the ability to think and understand things, especially complex or abstract ideas. It can also refer to a person's mental capacity or intelligence. ("बुद्धि" चीजों को सोचने और समझने की क्षमता को संदर्भित करती है, विशेष रूप से जटिल या अमूर्त विचारों को। यह किसी व्यक्ति की मानसिक क्षमता या बुद्धि का भी उल्लेख कर सकता है।)
 - **Example:** "She has a keen *intellect* and was able to understand the complex theory quickly."

Let's see the meanings of the given words

- "Genius" refers to a person who is exceptionally intelligent or talented, often used to describe someone who is a prodigy or has exceptional abilities in a specific field. ("जीनियस" एक ऐसे व्यक्ति को संदर्भित करता है जो असाधारण रूप से बुद्धिमान या प्रतिभाशाली है, अक्सर किसी ऐसे व्यक्ति का वर्णन करने के लिए उपयोग किया जाता है जो एक विलक्षण है या किसी विशिष्ट क्षेत्र में असाधारण क्षमता रखता है।)
 - **Example:** "Albert Einstein is considered a *genius* in the field of physics."

- **"Wizard"** refers to a person who is skilled or expert in a particular field, often used to describe someone who is highly proficient in the use of a particular technology or software. It can also refer to a person who has magical powers. ("जादूगर" एक ऐसे व्यक्ति को संदर्भित करता है जो किसी विशेष क्षेत्र में कुशल या विशेषज्ञ है, अक्सर किसी ऐसे व्यक्ति का वर्णन करने के लिए प्रयोग किया जाता है जो किसी विशेष तकनीक या सॉफ्टवेयर के उपयोग में अत्यधिक कुशल होता है। यह किसी ऐसे व्यक्ति को भी संदर्भित कर सकता है जिसके पास जादुई शक्तियाँ हों।)
 - **Example:** *"He is a **wizard** at programming and can fix any problem within minutes."*
- **"Stupidity"** refers to the state or quality of being stupid, which means lacking intelligence or common sense. ("मूर्खता" मूर्ख होने की स्थिति या गुणवत्ता को संदर्भित करता है, जिसका अर्थ है बुद्धि या सामान्य ज्ञान की कमी।)
 - **Example:** *"The **stupidity** of his decision led to the failure of the project."*
- **"Sage"** refers to a person who is wise and has a lot of knowledge and experience, often used to describe someone who is respected for their wisdom and insight. ("ऋषि" एक ऐसे व्यक्ति को संदर्भित करता है जो बुद्धिमान है और जिसके पास बहुत ज्ञान और अनुभव है, अक्सर किसी ऐसे व्यक्ति का वर्णन करने के लिए प्रयोग किया जाता है जो उनकी बुद्धि और अंतर्दृष्टि के लिए सम्मानित होता है।)
 - **Example:** *"The old **sage** had lived a long life and had much wisdom to impart to the young people."*

According to the meanings of the given words, the correct answer is **Option 3, i.e., Stupidity.**

Question 20

Directions: Read the following passage and answer the given questions:

Religion and science have often been viewed as having conflicting ideologies, with one relying on faith and the other on evidence. However, many argue that these two concepts can coexist and even complement each other. Religion can provide a moral framework and a sense of purpose, while science can explain the natural world and how it operates. One way in which religion and science can coexist is through the concept of intellect. Religion provides a moral framework and a sense of purpose, while science explains the natural world and how it operates. One example of this coexistence is in the study of the origins of the universe. Many religions have creation myths that describe how the world came to be, while science has the Big Bang theory. Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other.

In conclusion, religion and science are not mutually exclusive. They can coexist and even complement one another, providing a deeper understanding of the world and our place in it. Both offer unique perspectives and insights, and neither necessarily contradicts the other. The key is to approach each with an open mind and a willingness to consider different perspectives.

Give an example of how religion and science coexist in the study of the origins of the universe.

Options:

A. Many religions have creation myths, while science has the Big Bang theory.

B. Scientists have used telescopes and other observational techniques to study the history of the universe, while theologians have used scriptures.

- C. The universe is believed to have been created by the god Brahma.
- D. The big bang theory is a scientific explanation for the origins of the universe.

Answer: A

Solution:

The correct answer is "**Many religions have creation myths, while science has the Big Bang theory.**"

Key Points

- Option '1' is correct.
 - From the passage: '*.... **Many religions have creation myths that describe how the world came to be, while science has the Big Bang theory.** Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other.....*'
 - The passage states that many religions have creation myths that describe how the world came to be, while science has the Big Bang theory. Both perspectives offer unique insights and understandings of the origins of the universe, and neither necessarily contradicts the other.
 - Thus, in this context, we can say that the correct answer is option "1"
- Options 2, 3, and 4 are incorrect because there isn't sufficient supportive information in the passage to consider them to be the right answers.

Hence, the correct answer is "*Many religions have creation myths, while science has the Big Bang theory.*"

RAGA

Question 21

Recently in July 2022, who amongst the following has won the Wimbledon men's title 2022?

Options:

- A. Novak Djokovic
- B. Nick Kyrgios
- C. Bjorn Borg
- D. Roger Federer

Answer: A

Solution:

The correct answer is ‘**Novak Djokovic**’

Key Points

- Serbia Novak Djokovic won a seventh Wimbledon men’s title with a four-set triumph over Nick Kyrgios.
 - The top seed at the Championships has now won 21 major titles.
 - He is just one short of all-time record men’s leader Rafael Nadal, and one ahead of 20-time Grand Slam champion Roger Federer.
 - Roger Federer has won a men’s record eight titles at Wimbledon.
 - Djokovic joins Federer, Bjorn Borg and Pete Sampras as the only men’s players to have won four consecutive Wimbledon titles.
 - Kazakhstan’s Elena Rybakina made a brilliant comeback after being set down to beat Tunisian third-seed Ons Jabeur 3-6, 6-2, 6-2 and win the Women’s Singles title at Wimbledon.
 - Rybakina became the first-ever Kazakh female player in the Open Era to win a WTA title on grass and is also the Kazakh female player with the most WTA-level titles in the Open Era (3).
 - The 23-year-old also became the youngest female player to win the title in Wimbledon since Petra Kvitová in 2011.
 - The Wimbledon Championships 2022 is a Grand Slam tennis tournament that takes place at the All England Lawn Tennis and Croquet Club in Wimbledon, London, United Kingdom.
 - The tournament is played on grass courts, with all main draw matches to be played at the All England lawn tennis and croquet club, Wimbledon.
-

Question 22

Which movie has won the Best Feature Film award at the 68th National Film Awards announced in July 2022?

Options:

- A. Tanhaji
- B. KGF Chapter-1
- C. Badhaai Do
- D. Soorarai Pottru

Answer: D

Solution:

The correct answer is **Soorarai Pottru**.

Key Points

- The winners of the **68th National Film Awards-2020** were announced by Information & Broadcasting Ministry on 22 July 2022 in New Delhi.
- **Best Feature Film:** Soorai Potru
- **Best Actor:** Suriya for Soorai Potru & Ajay Devgn for Tanhaji
- Madhya Pradesh won the 'Most Film Friendly State' award.
- **Best Actress:** Aparna Balamurali for Soorai Potru.

Important Points

- The National Film Awards is the most prominent film award ceremony in India.
 - Established in 1954, it has been administered, along with the International Film Festival of India and the Indian Panorama, by the Indian government's Directorate of Film Festivals since 1973.
 - **The National Film Awards:**
 - First awarded: 10 October 1954; 67 years ago
 - Location: Vigyan Bhavan, New Delhi
 - Presented by: Directorate of Film Festivals
-

Question 23

Which Indian state is the largest by area?

Options:

- A. Maharashtra
- B. Uttar Pradesh
- C. Rajasthan
- D. Madhya Pradesh

Answer: C

Solution:

The correct answer is Rajasthan.

Key Points

- **Rajasthan is the largest state in India in terms of area.**
- It covers an area of **342,239 sq. km** and has a population of **6.85 crores** as per the 2011 census.
- **Uttar Pradesh** is the largest state of India in terms of population.

Additional Information

- Area-wise states-

S.No.	States
1	Rajasthan
2	Madhya Pradesh
3	Maharashtra
4	Uttar Pradesh

- Population-wise states-

S.No.	States
1	Uttar Pradesh
2	Maharashtra
3	Bihar
4	West Bengal

Question 24

In which medium sound travels faster?

Options:

- A. Solid
- B. Liquid
- C. Gas
- D. None of these

Answer: A

Solution:



Key Points

Sound waves:

- Sound waves are **mechanical waves**, that is, they require a **material medium to travel**.
- It is the movement of the medium that transmits the wave.
- Since the **density of solids is high and the molecules of a solid are closely packed**, sound waves are **quickly transmitted from one molecule to the next** thus enabling it to travel faster.

- The **speed of sound**, therefore, is **fastest in solids**, followed by liquids (less dense than solids) and then gas (least dense).
- **Sound cannot travel in a vacuum** as there is no material medium.

★ Important Points

Speed of Sound in Different Media:

- The speed of sound in the **air** is **331 m/s⁻¹ at 0°C** and **344 m/s⁻¹ at 22°C**.
- It depends on the **type of medium through which it is travelling**.
- It depends on the **temperature and pressure** of that medium.
- Speed of sound **Decrease**-when it **passes from the solid to the gaseous state** of a given medium.
- Speed of sound **Increase**- In any medium, when **temperature and humidity increase**.
- **Speed of Sound- Solid > Liquid > Gases**

Thus, in solid medium sound travels faster.

Question 25

Who is the current President of India?

Options:

- A. Ram Nath Kovind
- B. Droupadi Murmu
- C. Jagdeep Dhankhar
- D. M. Venkaiah Naidu

Answer: B

Solution:

The correct answer is **Droupadi Murmu**.



Key Points

- **Droupadi Murmu**
 - She was the **Former governor of Jharkhand and tribal leader from Odisha**, Droupadi Murmu.
 - She was the first member of a scheduled tribe to be nominated for the position of **President of India**.
 - The **BJP-led NDA** officially nominated her for the **2022 presidential race**.
 - Droupadi Murmu is serving as **India's 15th president**.
 - She triumphed over **Yashwant Sinha**, the joint candidate for the opposition.



Additional Information

- She is the first President born in Independent India and the **youngest person to hold the position**.
 - Before being chosen as president, she served in a **number of cabinet positions for the Government of Odisha** between 2000 and 2004.
 - She was the **ninth governor of Jharkhand** from 2015 to 2021.
-

Question 26

Where was the capital of Shivaji Maharaj?

Options:

- A. Raigarh
- B. Sindhudurg
- C. Poona
- D. Kolhapur

Answer: A

Solution:

The correct answer is **Raigarh**.

Key Points

- **Raigarh** was the capital of Shivaji.
 - **Shivaji:**
 - **Chhatrapati Shivaji Maharaj**, great **Maratha ruler** was born on **19 February 1630** at **Shivneri Fort** in **Pune District** in the present-day state of Maharashtra.
 - **Shivaji** was given excellent training in **military warfare and administration**.
 - He was married for the first time in **1640 to Saibai**.
 - **Shivaji** displayed his military zeal for the first time in **1645 when as a teenager**.
 - He successfully got control of the **Torna Fort which was under Bijapur**.
 - He also acquired the **Kondana Fort**.
 - Both these forts were under **Adil Shah of Bijapur**.
 - Shah then got Shivaji imprisoned in a bid to contain Shivaji.
 - Some accounts say that Shivaji surrendered these forts to get his father released.
 - **Shahaji died in 1664-65** in an accident.
 - After this, **Shivaji** resumed his raids and extended his territories.
-

Question 27

How many spokes are there in the Ashoka Chakra depicted on the National flag of India?

Options:

A. 36

B. 24

C. 16

D. 32

Answer: B

Solution:

The Correct Answer is 24.

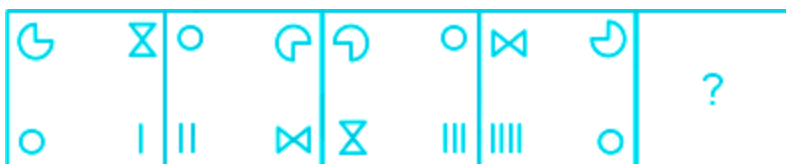
- The Ashoka Chakra which features in the middle of the tricolour, has **twenty-four spokes**.
- It was adopted on July 22, 1947.

★ Important Points

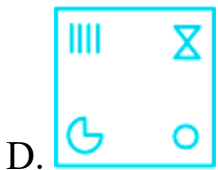
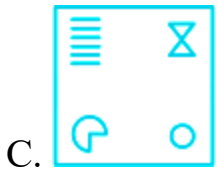
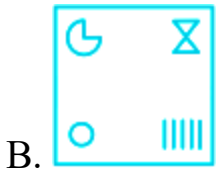
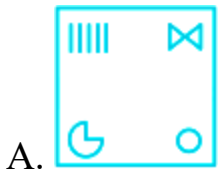
- The wheel is called the Ashoka Chakra because it appears on a number of edicts of Ashoka.
 - Each spoke on the chakra symbolizes one principle of life and also the twenty-four hours in the day, which is why it is also called the '**Wheel of Time**'.
 - Some of the twenty-four principles that the spokes represent are — **love, courage, patience, self-sacrifice, truthfulness, righteousness, spiritual knowledge, and faith, among others.**
-

Question 28

Select the figure that will come in place of the question mark (?) in the following figure series.



Options:



Answer: B

Solution:

Explanation:

The pattern followed in the given series is:

The number of lines increases in every step and it moves left to right in the bottom.

The partial circle moves left to right in the top and it moves clockwise.

The ring moves clockwise.

Next step:

The number of lines will be 5.

The partial circle will be at left and face towards right.

The ring will be at bottom left.

Now we consider the given options:

Option 1) The partial circle is placed incorrectly.

Option 2) The pattern is correct.

Option 3) The partial circle is placed incorrectly.

Option 4) The partial circle is placed incorrectly.

Hence the correct answer is option 2.

Question 29

Select the figure that does not belong in the following series.



Options:

A. A

B. D

C. B

D. C

Answer: C

Solution:

The logic is followed here :



A

⇒ This diagram contains four elements , , , 



C

⇒ This diagram contains four elements , , , 



D

⇒ This diagram contains four elements , , , 



B

⇒ This diagram **does not contain** same four elements.  is missing from the diagram.

So, B is not belong to the series.

Hence, the correct answer is "option 2".

Question 30

Five girls are sitting in a row. Rashi is not adjacent to Sulekha or Abha. Anuradha is not adjacent to Sulekha. Rashi is adjacent to Monika. Monika is at the middle of the row. Then, Anuradha is adjacent to whom out of the following?

Options:

A. Rashi

B. Abha

C. Monika

D. Sulekha

Answer: A

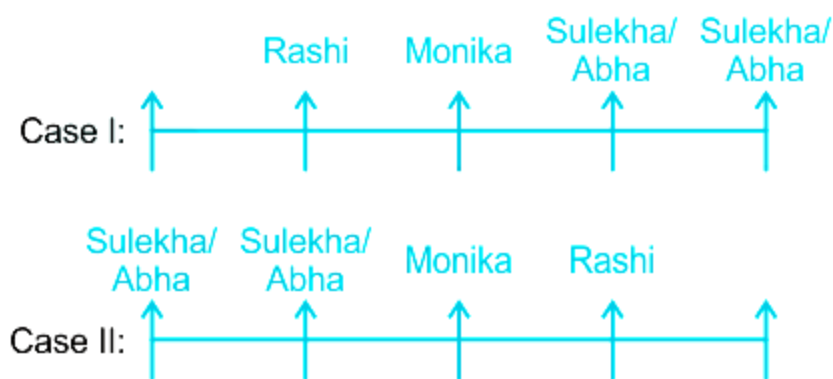
Solution:

1. Rashi is adjacent to Monika.

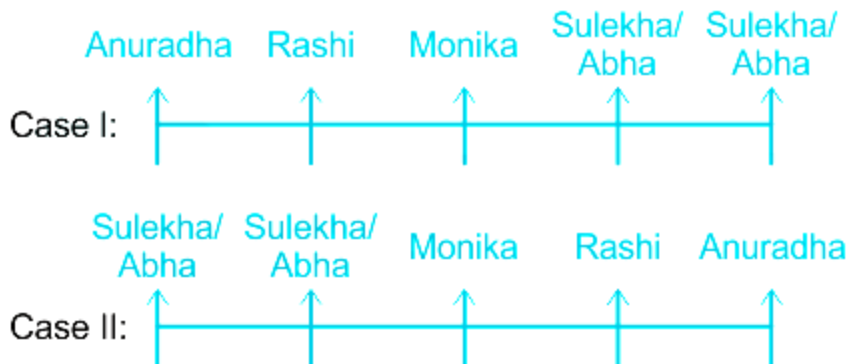
2. Monika is at the middle of the row.



3. Rashi is not adjacent to Sulekha or Abha.



4. Anuradha is not adjacent to Sulekha.



Clearly, Anuradha is adjacent to Rashmi.

Hence, '**Rashmi**' is the correct answer.

Question 31

If a mirror is placed on the shaded line, then which of the following options is the mirror image of the given figure?

74sJKt26

Options:

A. 74sJKt26

B. 74sJKt26

C. 95tJKs75

D. 74sJKt26

Answer: D

Solution:

The logic follows here is:

When the mirror is placed on the shaded line, the left side and right side changed vice-versa where the top and bottom remain the same.

Thus, the correct mirror image of the given figure will be:

74sJKt26 | 74sJKt26

Hence, the correct answer is "Option 4".

Additional Information

In Mirror image left side and right side changed vice-versa where top and bottom remain the same.

Reflection of an object into the water is its water image. It appears by inverting an object vertically i.e. upside down.

The water image of the figure looks like mirror image of the figure in case the mirror is horizontally at the bottom of the figure.

Water image is just a reflection where top and bottom part of the images changed where left and right side of the image remain same.

Question 32

A paper is folded and cut as shown below. How will it appear when unfolded?



Options:



A.



B.



C.

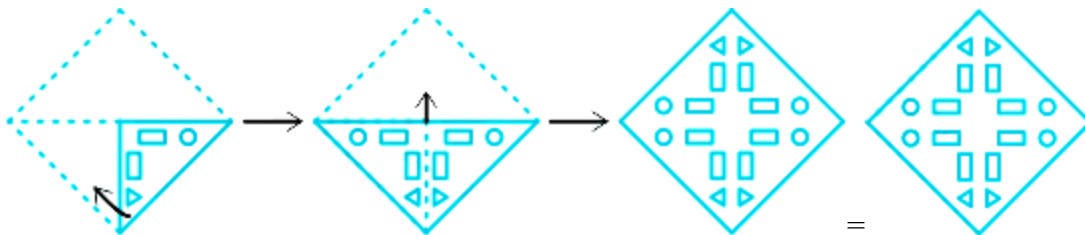


D.

Answer: B

Solution:

Here we can see that the given paper is folded three times.



Hence, the correct answer is "**Option2**"

Question 33

Pointing to a girl in a photo, Amitha said, "She is the sister of the son of the wife of my husband."

How is the girl in photo related to Amitha?

Options:

- A. Sister
- B. Daughter
- C. Niece
- D. Sister-in-law

Answer: B

Solution:

Preparing the family tree using the following symbols,

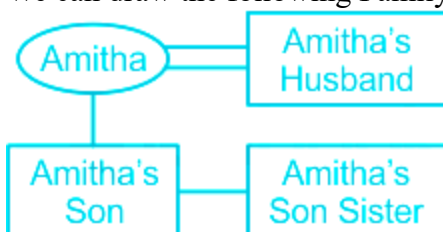
Symbol in Diagram	Meaning
○	Female
□	Male
==	Married Couple
—	Siblings
	Difference of A Generation

Given:

"She is the sister of the son of the wife of my husband."

Break the statement: Amithahusband's wife is Amitha.

We can draw the following Family Tree from the information given above:



Hence, the girl is the "**Daughter**" of Amitha

Question 34

A lion prowled 900 m North from its den. It took a right turn and prowled 800 m. It, then took a right turn and prowled 700 m. It, again, took a right turn and prowled 600 m. Which direction is the lion facing now?

Options:

A. North

B. West

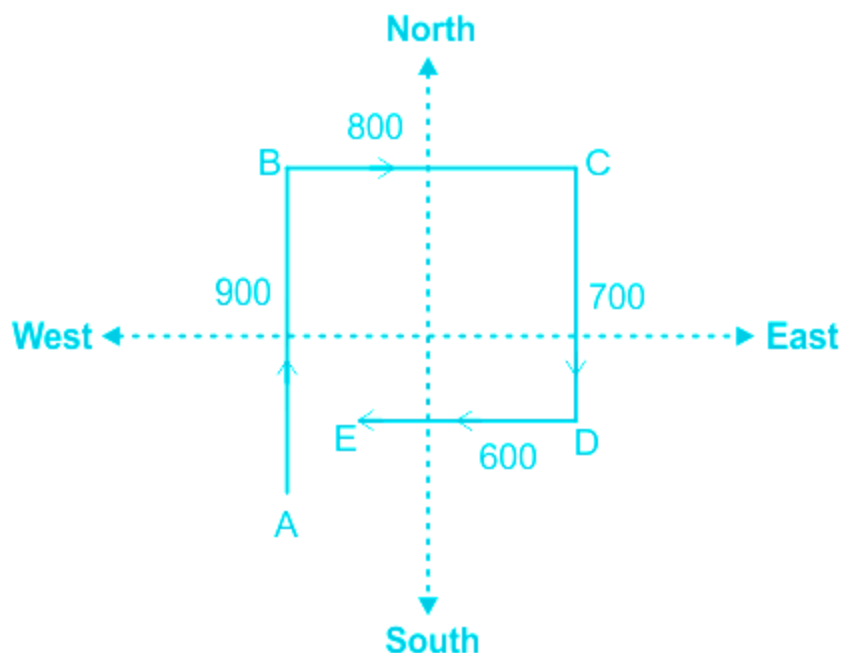
C. East

D. South

Answer: B

Solution:

As per the given information, the direction and distance diagram is as follows:



At the end, the lion is facing in the **West** direction.

Hence, the correct answer is "West".

Question 35

Shweta is taller than Anu and Vani is smaller than Dev, Tanu is Taller than Vani and smaller than Samar, Anu is taller than Samar. Dev is smaller than Anu.

Who is tallest among all?

Options:

A. Shweta

B. Vani

C. Anu

D. Samar

Answer: A

Solution:

Arrangement of heights according to the given information:

(1) Shweta is taller than Anu.

$\text{Shweta} > \text{Anu}$

(2) Vani is smaller than Dev.

$\text{Dev} > \text{Vani}$

(3) Tanu is Taller than Vani and smaller than Samar.

$\text{Samar} > \text{Tanu} > \text{Vani}$

(4) Anu is taller than Samar.

$\text{Anu} > \text{Samar}$

(5) Dev is smaller than Anu.

$\text{Anu} > \text{Dev}$

Possible arrangement:

Shweta > Anu > Dev > Samar > Tanu > Vani

The position of Dev may vary with Samar and Tanu.

Here, **Shweta** is the tallest among all.

Here, the correct answer is "**Shweta**".

Question 36

What should come in the place of question mark (?) in the following?

Maharashtra : India :: Texas : ?

Options:

A. USA

B. Canada

C. Mexico

D. Turkey

Answer: A

Solution:

The logic followed here is:

Maharashtra : India → Maharashtra is the state of India.

Similarly,

Texas : USA → Texas is the state of the USA.

Hence, the correct answer is "**USA**".

Question 37

Which of the following represent the correct relationship between Cat, Dog, Pet ?

Options:

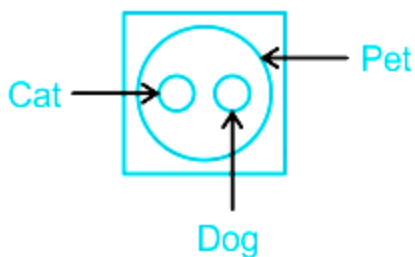


Answer: B

Solution:

The logic followed here is:

Dog and Cat are the Pet animals.



The correct answer is "option (2)".

Question 38

In a certain code language, if PERSON is coded as "OCOOJH", then how is REPORT coded in that language?

Options:

- A. QCMKMN
- B. QCMKMM
- C. OCMKMN
- D. OCMKMM

Answer: A

Solution:

The logic followed here is:

P	E	R	S	O	N
-1	-2	-3	-4	-5	-6
↓	↓	↓	↓	↓	↓
O	C	O	O	J	H

Similarly,

R	E	P	O	R	T
-1	-2	-3	-4	-5	-6
↓	↓	↓	↓	↓	↓
Q	C	M	K	M	N

Hence, the correct answer is "**option (1)**".

Question 39

Find the mirror image of 8:20 ?

Options:

- A. 3:20
- B. 3:40
- C. 3:30
- D. 4:40

Answer: B

Solution:

While using the formula:

Mirror Image of Clock = 11:60 - Given Time

$$= 11:60 - 8:20$$

$$= 3:40$$

Hence, the correct answer is **"option (2)"**.

Question 40

Find the missing number.

0, 3, 8, 15, 24, ? 48

Options:

A. 35

B. 30

C. 27

D. 36

Answer: A

Solution:

The logic followed here is:



The correct answer is **"option (1)"**.

Question 41

Select the option that is related to the third figure in the same way as the second figure is related to the first figure.



Options:



Answer: C

Solution:

The logic followed in the given figures:



Figure (2) is the water image of figure (1).



Similarly,



Figure (4) is the water image of figure (3).



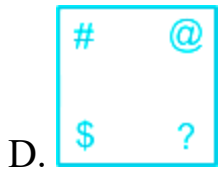
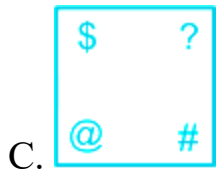
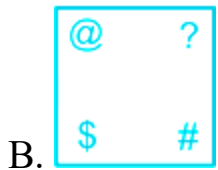
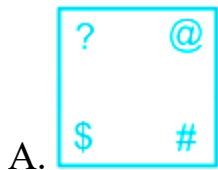
Hence, the correct answer is "Option (3)".

Question 42

Select the figure from the options that can replace the question mark (?) in the following figure series.



Options:



Answer: C

Solution:

The images in the figure are shifting its position by rotating in:

- For '@'- It is shifting in the anti-clockwise direction.



- For '\$'- It is shifting in the anti-clockwise direction.



- For '#'- It is shifting in the anti-clockwise direction.



- For '?'- It is shifting in the anti-clockwise direction.

\$ ← ?	? ↓ #	# @	@ \$	\$?
@ #	\$ @	? → \$	# ?	@ #

- For 'LINE' (**I**) -It is not shifting and are in the decreasing order.

\$?	? #	# @	@ \$	\$?
@ #	\$ @	? \$	# ?	@ #

Hence, "Option - (3)" is the correct answer.

Question 43

If $[(216)^4 \div 36^4] \times 6^5 = 6^m$ then $m = ?$

Options:

- A. 13
- B. 11
- C. 10
- D. 9

Answer: D

Solution:

Formula used:

- $(a^m)^n = a^{mn}$
- $a^m \times a^n = a^{m+n}$
- $a^m / a^n = a^{m-n}$

Calculation:

$$[(216)^4 \div 36^4] \times 6^5 = 6^m$$

$$\Rightarrow [(6^3)^4 \div (6^2)^4] \times 6^5 = 6^m$$

By using the above properties

$$\Rightarrow [(6^{12-8}) \times 6^5 = 6^m]$$

$$\Rightarrow 6^4 \times 6^5 = 6^m$$

$$\Rightarrow 6^9 = 6^m$$

$$\therefore m = 9$$

Question 44

The perimeter of a rectangle and a square are 160 m each. The area of the rectangle is less than that of the square by 100 sq. m. The length of the larger side of the rectangle is :

Options:

A. 70 m

B. 60 m

C. 40 m

D. 50 m

Answer: D

Solution:

Formula Used:

$$\text{Perimeter of rectangle} = 2(l + b)$$

$$\text{Area of rectangle} = l \times b$$

$$\text{Perimeter of square} = 4a$$

$$\text{Area of square} = a^2$$

Where,

l & b are length & breadth of rectangle

a is the side of square

Calculation:

According to the question

$$2(l + b) = 160$$

$$l + b = 80$$

$$l = 80 - b \text{ -----(1)}$$

Also,

$$4a = 160$$

$$\mathbf{a = 40}$$

Again, according to the question

$$\Rightarrow a^2 - (l \times b) = 100$$

$$\Rightarrow 1600 - lb = 100$$

$$\Rightarrow \mathbf{lb = 1500}$$

$$\Rightarrow (80 - b)b = 1500 \text{ [from equation (1)]}$$

$$\Rightarrow 80b - b^2 = 1500$$

$$\Rightarrow b^2 - 80b + 1500 = 0$$

$$\Rightarrow b^2 - 50b - 30b + 1500 = 0$$

$$\Rightarrow b(b - 50) - 30(b - 50) = 0$$

$$\Rightarrow (b - 50)(b - 30) = 0$$

$$\Rightarrow b = 50, 30$$

$\therefore$ The length of the rectangle is 50 cm.

Question 45

If diagonal of cube is $\sqrt{12}$ cm, then its volume in cm^3 is:

Options:

A. 18

B. 9

C. 6

D. 8

Answer: D

Solution:

Given:

Diagonal of cube $= \sqrt{12}$

Formula used:

Volume of cube $= (\text{side})^3$

Volume of diagonal $= \text{side} \times \sqrt{3}$

Calculation:

According to the question:

Volume of diagonal $= \text{side} \times \sqrt{3}$

$\Rightarrow \sqrt{12} = \text{side} \times \sqrt{3}$

$\Rightarrow \sqrt{(4 \times 3)} = \text{side} \times \sqrt{3}$

$\Rightarrow 2\sqrt{3} = \text{side} \times \sqrt{3}$

$\Rightarrow \text{side} = 2 \text{ cm}$

Now,

Volume of cube $= (\text{side})^3$

$\Rightarrow (2)^3 = 8 \text{ cm}^3$

$\therefore$ The volume of a cube is 8 cm^3 .

Question 46

In an election between two candidates, one got 56% of the total valid votes, 14% of the votes are invalid. If the total number of votes was 90000, what was the number of valid votes the other candidate got?

Options:

A. 42990

B. 43344

C. 52345

D. 34056

Answer: D

Solution:

Given:

one got 56% of the total valid votes

14% of the votes are invalid

If the total number of votes was 90000

Calculation:

Total number of votes = 90000

14% of the votes are invalid,

Hence, (100% - 14%) of the votes are valid

Valid votes = 86% of 90000

$$= \frac{86 \times 90000}{100}$$

$$= 77400 \text{ votes}$$

One got 56% of the valid votes

$$= 56\% \text{ of } 77400$$

$$= \frac{56 \times 77400}{100}$$

$$= 43344 \text{ votes}$$

Number of the valid votes of the other candidate = 77400 - 43344

$$= 34056 \text{ votes}$$

$\therefore$ Answer is 34056.

Question 47

A man walks at the rate of 5 km/hr for 6 hours and at 4 km/hr for 12 hours. The average speed of the man is:

Options:

A. 4

B. $4\frac{1}{2}$

C. $4\frac{1}{3}$

D. $4\frac{2}{3}$

Answer: C

Solution:

Formula used:

Average speed = Total Distance / Total time

Calculation:

Case-1 Person walks at the rate of 5 km/hr for 6 hours

Distance = $5 \times 6 = 30$ km

Case-2 Person walks at the rate of 4 km/hr for 12 hours

Distance = $4 \times 12 = 48$ km

Average speed = Total Distance / Total time

Average Speed = $\frac{30+48}{6+12} = \frac{78}{18} = 4\frac{1}{3}$

$\therefore$ The correct answer is $4\frac{1}{3}$.

Question 48

A shopkeeper sells his articles at their cost price but uses a faulty balance which reads 1000 gm for 800 gm. What is the actual profit percentage?

Options:

A. 20%

B. 25%

C. 30%

D. 40%

Answer: B

Solution:



Shortcut Trick

Let CP = SP = 100

$$\begin{array}{rcl} \text{CP} & : & \text{SP} \\ 100 & : & 100 \\ 4800 & : & 10005 \\ \hline 4 & : & 5 \end{array}$$

$$\Rightarrow \text{Profit Percentage} = (1/4) \times 100 = 25\%$$

$\therefore$ The actual profit percentage is 25%.

Given:

Actual weight of article = 1000 gm

Faulty weight of article = 800 gm

Formula used:

$$\text{Profit percent due to wrong weight} = [\text{Error}/(\text{True weight} - \text{Error})] \times 100$$

Calculation:

Actual weight of article = 1000 gm

Faulty weight of article = 800 gm

$$\text{Error} = 1000 - 800 = 200 \text{ gm}$$

$$\text{Profit percent due to wrong weight} = [\text{Error}/(\text{True weight} - \text{Error})] \times 100$$

$$\Rightarrow (200/800) \times 100 = 25\%$$

$\therefore$ The actual profit percentage is 25%.

Question 49

The LCM of two numbers is 48. The numbers are in the ratio 2:3. The sum of the numbers is -

Options:

A. 28

B. 32

C. 40

D. 64

Answer: C

Solution:

Given

The LCM of two numbers is 48. The numbers are in the ratio 2:3.

Concept used

Lcm and Hcf

Calculation

Let two numbers be $2x$ and $3x$

Their Lcm = $6x$

$$\Rightarrow 6x = 48$$

$$\Rightarrow x = 8$$

Sum of the numbers

$$\Rightarrow 2x + 3x = 5x$$

$$\therefore 5 \times 8 = 40$$

Question 50

After taking 1/5th of the contents out of my purse, 1/12th of the remainder was found to be Rs. 7.40. How much money did the purse initially contain?

Options:

A. Rs. 115

B. Rs. 113

C. Rs.111

D. Rs. 106

Answer: C

Solution:

Calculation:

Let, the purse contained X rupees initially.

$$\Rightarrow \frac{1}{12} \times \left[1 - \frac{1}{5}\right] \times X = 7.40$$

$$\Rightarrow \frac{1}{12} \times \frac{4}{5} \times X = 7.40$$

$$\Rightarrow (X/15) = 7.40$$

$$\Rightarrow X = 111$$

The purse initially contain Rs. 111
